FINAL YEAR PROJECT / BENG (HONS) ELECTRONICS & COMPUTER ENGINEERING

Smart Walking Cane with Vision-Based Perception & Finger-Allocated Haptic Feedback

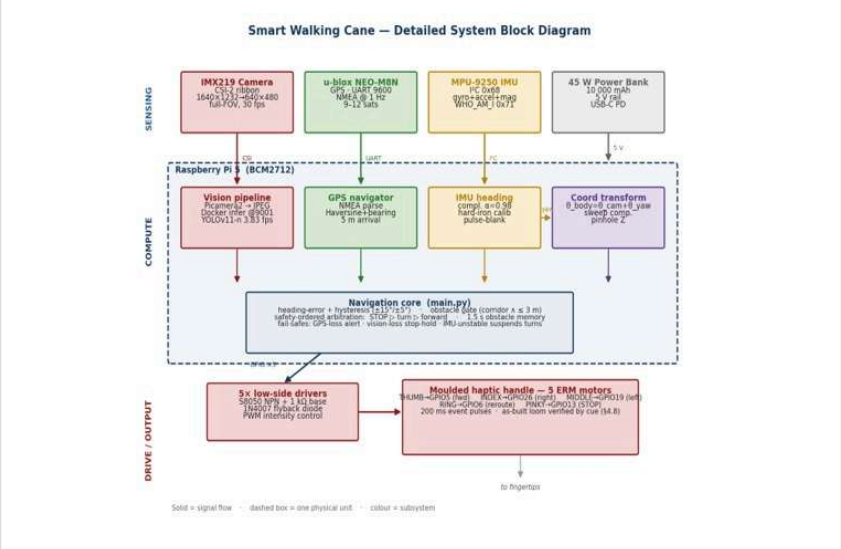
IN ACADEMIC COLLABORATION WITH A·P·U ASIA PACIFIC UNIVERSITY OF TECHNOLOGY & INNOVATION

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Visually impaired pedestrians rely on audio cues that **mask the environmental hearing** they need to stay safe. This cane moves directional guidance onto the skin — **five vibration motors, one per finger** — leaving hearing entirely free.

SYSTEM OVERVIEW



Raspberry Pi 5 fuses on-device vision, GPS and IMU heading to drive five finger motors — fully offline.

AIM & OBJECTIVES

01 Analyse limits of audio and single-motor feedback.

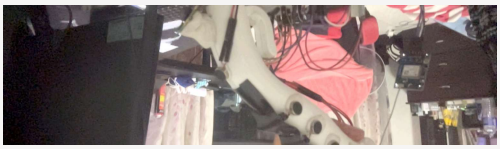
02 Build a finger-allocated haptic system in a moulded handle.

03 Assess effectiveness through controlled testing.

FINGER → CUE MAP

Thumb	Forward
Index	Right
Middle	Left
Ring	Reroute
Pinky	Stop

MOULDED HANDLE · 5 ERM MOTORS



METHODOLOGY

On-device inference
Containerised YOLOv11-nano, fully offline — privacy preserved.

Measured geometry
Camera height 1.03 m, tilt 13.5° for range estimation.

Event-driven haptics
200 ms pulses fired only on a directional decision.

Three-tier testing
Self-test → deterministic simulation → live hardware.

KEY RESULTS

3.83 fps
On-device YOLOv11-nano — **8.5× faster** than the transformer baseline.

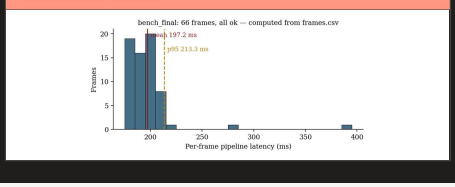
60.3%
mAP@50 validation accuracy.

73%
Haptic localisation — up from **57%**.

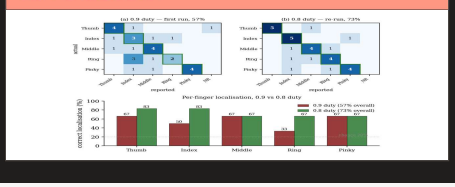
±3°
Heading accuracy after calibration.

RM 894
Unit cost vs **RM 2,000–8,000** commercial.

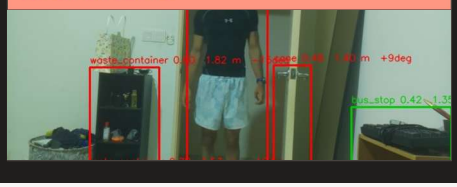
INFERENCE SPEED



HAPTIC ACCURACY



DETECTION SAMPLE



SUSTAINABLE DEVELOPMENT GOALS

SDG 10 — **Reduced Inequalities**
Assistive capability at a fraction of commercial cost.

SDG 11 — **Sustainable Cities & Communities**
Independent access to public space and transport.

CONCLUSION

An integrated, low-cost cane that **relocates directional guidance to the fingertips** — validated across its vision, heading and haptic subsystems. On-device perception preserves both **privacy and offline function**, keeping the user's hearing free for the world it was meant to sense.